Saket Mehla

Roll No.: 230103087
B.Tech - Mechanical Engineering
Indian Institute of Technology Guwahati

+91-8398856425
m.saket@iitg.ac.in
saketmehla@gmail.com
saket-mehla | m-saket | Portfolio

Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech	Indian Institute of Technology Guwahati	7.44	2023-Present
Senior Secondary	CBSE Board	95.6%	2023
Secondary	CBSE Board	99.8%	2021

Experience

- **American Express** *May 2026 - Jul 2026*
  Analyst Intern | *Letter of Recommendation* Gurgaon, India
  - Advanced **Pre-Auth fraud** decisioning for token provisioning, shifting controls upstream for **30%**+ AmEx **payment volume**
  - Mapped **300**+ **baseline** non-transactional **features** across fraud-risk themes to uncover gaps in existing model intelligence
  - Augmented the stack with **23** orthogonal, **point-in-time-safe variables** across real-time, linkage, channel and account risk
  - Validated feature stack with **5% overall** and 10% segment-level **fraud-capture lift**, yielding **$1.5M** projected annual savings

Projects

- **Credit Risk Behaviour Scoring using Ensemble Models** | IDFC FIRST Bank *Jan 2025 - Feb 2025*
  *Convolve 3.0 – **Inter IIT ML/AI Hackathon*** bit.ly/convolve-idfc
  - Analyzed 96K+ accounts, screening 1.2K+ predictors through **MI, ANOVA, AUC & KS** to retain top 200 risk features
  - Evaluated PCA, resampling & cost-sensitive learning under **5-fold stratified OOF** to model a 1.42% rare-default cohort
  - Blended **XGBoost & CatBoost** via OOF weighting, achieving **0.857 ROC-AUC** & capturing **39.8%** of **defaults** in top 5%
- **Chatter Detection Model** | Course Project *Mar 2026 - Apr 2026*
  Prof. RK Mital bit.ly/chatter-model
  - Built **engagement detection** on 3-axis, 10 kHz raw force signals to isolate genuine cutting segments for feature extraction
  - Designed a **frequency-ratio chatter label** (FFT peak vs. harmonic amplitude, $>$**10**$\times$ threshold) with persistence filtering
  - Benchmarked **8 classifiers** (Random Forest, XGBoost, SVM) via GroupKFold CV, achieving **0.97 recall** and **0.95 F2-score**
- **Solar Capacity & Waste Forecasting** | INA Solar *Nov 2024 - Dec 2024*
  *Inter IIT Tech Meet 13.0, IIT Bombay – **Bronze Medal*** bit.ly/INA-Solar
  - Forecasted India's future solar installation capacity using 15Y data via degree-optimized **polynomial regression R^2=0.996**
  - Simulated waste via parameterized **Weibull distribution** (α=**5.37**) & EOL degradation, reaching **16.68M tonnes** by 2050
  - Analyzed state-wise solar capacity distribution, identifying **9** states with **90% capacity** as high-priority recycling zones
- **Memory-Augmented LLM Engine using Hybrid RAG** | AI Mirage *Sep 2025 - Oct 2025*
  Inter IIT Tech Bootcamp bit.ly/AI-Mirage
  - Built a **multi-model LLM engine** orchestrating **6**+ **task-specific workflows** across Groq/Gemini with Llama Guard safety
  - Designed **hybrid memory** with 5-turn summaries & **2 ChromaDB** stores, retrieving top-3 events through **semantic RAG**
  - Automated QA across memory, NPC & quest states; passed **4/4 recall tests** while sustaining context over **30**+ **chat turns**

Technical Skills

- **Languages & Querying:** Python, C, C++, SQL
- **Databases:** Google BigQuery, PostgreSQL, MySQL
- **Data Science:** NumPy, Pandas, Hypothesis Testing, Scikit-learn, PyTorch, TensorFlow, Clustering, Ensemble Learning

Key Courses Taken

- **Mathematics:** Mathematics I/II/III (Multivariable Calculus, Linear Algebra, ODE, PDE, Fourier & Laplace Transforms)
- **Data & Analytics:** Machine Learning for Engineers, Industrial Engineering & Operations Research, Experimental Methods

Achievements & Extracurriculars

- **Bronze Medal**, Inter IIT Tech Meet 13.0, IIT Bombay – Energy Consulting Problem Statement among all **23 IITs** *2024*
- **Global Rank 3**, Strategy Storm, IIT Guwahati – Global Business Case Competition among **3,700**+ **teams** *2025*
- **Top 18**, Indian Case Challenge, IIT Kharagpur – Business Strategy Case Competition among **5,300**+ **teams** *2025*
- **2**$\times$ **Gold Medals**, Inter-Hostel Tech Competition – EcoEdge & Water Wise Consulting Challenges among **14** hostels *2025*
- **Amazon ML Summer School** – Selected for Amazon MLSS, an ML program for the **top 5%** from 61,000+ applicants *2024*
- **Core Team Member**, Consulting & Analytics Club – Co-organized 6-week **Summer Analytics'25** for **15,000**+ **learners** *2025*
- **National Rank 2**, Secondary School Examination | **School Topper**, Senior Secondary School Examination *2021, 2023*
- **Bronze Medal**, National Fencing Championship, representing **Haryana** in the national-level competition *2019*

Positions of Responsibility

- **Marketing and Outreach Lead**, Inter IIT Tech Meet 15.0 *Apr 2026 - Present*
  - Spearheading IITG's **first** offline **Inter IIT Tech Meet**, aligning all **23** IIT Technical Boards for **2,000**+ onsite **participants**
  - Leading **3 Heads** & **40**+ **Core Members**, driving industry partnerships across **4 tracks** and 8+ diverse technology domains
  - Sourced **13**+ industry-sponsored **problem statements** worth **INR 50L**+, contributing to INR 2Cr+ overall event budget
- **Student Placement Coordinator**, Centre for Career Development, IIT Guwahati *Apr 2025 - Apr 2026*
  - Led end-to-end **campus hiring** for **2,500**+ students, enabling **350**+ internship and **1,000**+ full-time offers across programs
  - Managed slotting, **40**+ PPTs and **5**+ workshops; represented IITG in **5**+ **leadership discussions** with **corporate leaders**